

Initiating Search

November 10, 2025, 2:19 PM

• Search:

CAS Lexicon Search:

Tempo - Preferred Concept

Search Tasks

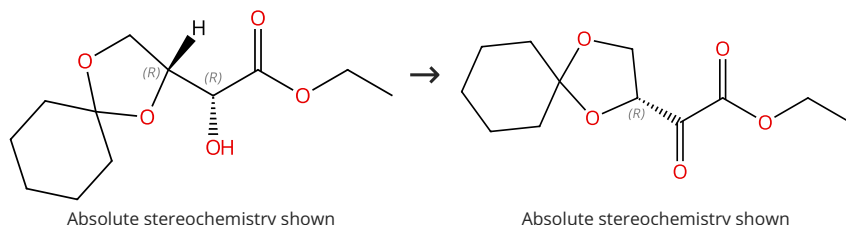
Task	Result Type	View
Returned Reference Results from CAS Lexicon (23,145)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,199)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	
Language:	English	
Publication Year:	2020 to 2025	

Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-31459235

Steps: 1 Yield: 100%

Toward the stereoselective synthesis of zaragozic acid framework: A desilylation-aldol reaction approach

By: Kurniawan, Yudhi D.; et al

Tetrahedron Letters (2021), 85, 153493.

1.1 Reagents: Trichloroisocyanuric acid

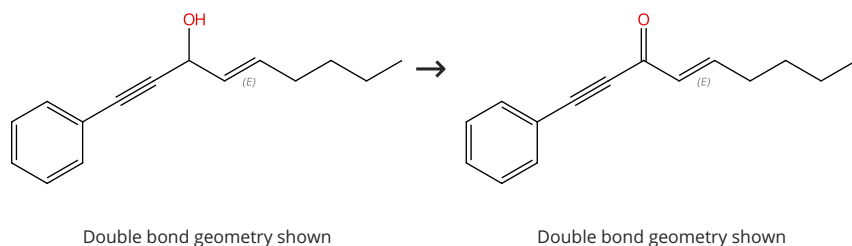
Catalysts: Tempo

Solvents: Dichloromethane; 2 h, rt

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



31-480-CAS-21503054

Steps: 1 Yield: 100%

A sustainable access to ynones through laccase/TEMPO-catalyzed metal- and halogen-free aerobic oxidation of propargylic alcohols in aqueous medium

By: Silva, Alana B. V.; et al

Catalysis Communications (2020), 137, 105946pp..

1.1 Reagents: Hydrochloric acid

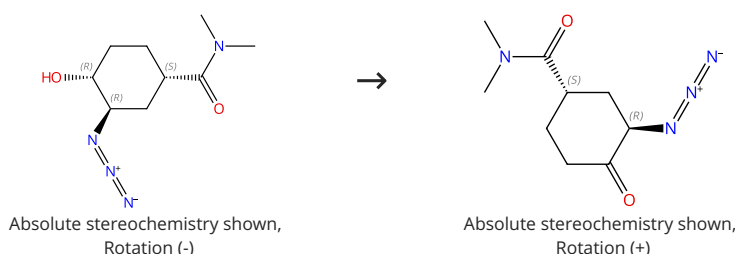
Catalysts: Tempo, Laccase

Solvents: Water; 8 h, pH 6, 30 °C

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-40647305

Steps: 1 Yield: 100%

A New Method to Large-Scalable Preparation of the Key Chiral Cyclohexane cis-Diamine Intermediate for Edoxaban

By: Yuan, Dingding; et al

Organic Process Research & Development (2024), 28(7), 2623-2634.

1.1 Reagents: Sodium bicarbonate, Tempo, Sodium hypochlorite,

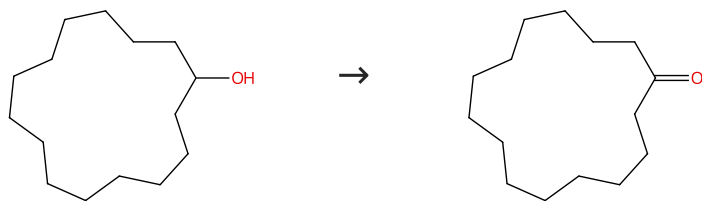
Potassium bromide

Solvents: Dichloromethane, Water; -5 °C → 5 °C

Experimental Protocols

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (52)

Suppliers (69)

31-480-CAS-22687563

Steps: 1 Yield: 100%

Photochemical transformations with iodine azide after release from an ion-exchange resin

By: Koesel, Teresa; et al

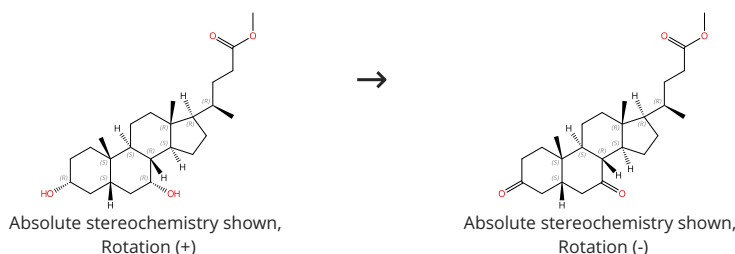
Angewandte Chemie, International Edition (2020), 59(30), 12376-12380.

- 1.1 **Reagents:** Azidotrimethylsilane (reaction products with Amberlyst A26), Lithium iodide (reaction products with Amberlyst A26), Amberlyst A 26 (reaction products with Li I, (diacetoxyiodo)benzene, trimethylsilyl azi...)
Solvents: Acetonitrile; 18 h, rt

Experimental Protocols

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (52)

Suppliers (11)

31-614-CAS-34334478

Steps: 1 Yield: 100%

Selectivity of Oxidizing Agents toward Axial and Equatorial Hydroxyl Groups

By: Kaspar, Miroslav; et al

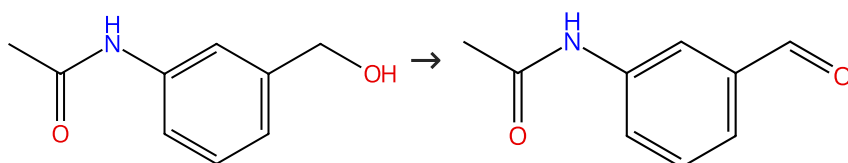
Journal of Organic Chemistry (2022), 87(14), 9157-9170.

- 1.1 **Reagents:** Chromium trioxide
Solvents: Acetone, Water; 0 °C; 30 min, 0 °C
- 1.2 **Reagents:** Isopropanol

Experimental Protocols

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (42)

Suppliers (44)

31-614-CAS-36103383

Steps: 1 Yield: 100%

Catalytic asymmetric C-C cross-couplings enabled by photoexcitation

By: Crisenza, Giacomo E. M.; et al

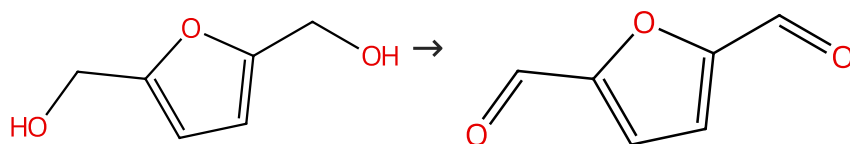
Nature Chemistry (2021), 13(6), 575-580.

- 1.1 **Reagents:** Pyridinium chlorochromate
Solvents: Dimethylformamide, Dichloromethane; 2 h, rt

Experimental Protocols

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

Suppliers (73)

31-614-CAS-41257728

Steps: 1 Yield: 100%

- 1.1 **Catalysts:** Copper bromide, 1,5-Naphthalenediamine, N^1, N^1, N^5, N^5 -tetrakis[[1-(phenylmethyl)-1*H*-1,2,3-triazol-4-yl]methyl]-
Solvents: Acetonitrile; 15 min, rt
- 1.2 **Catalysts:** 1-Methylimidazole, Tempo
Solvents: Acetonitrile; 2 h, rt

Experimental Protocols

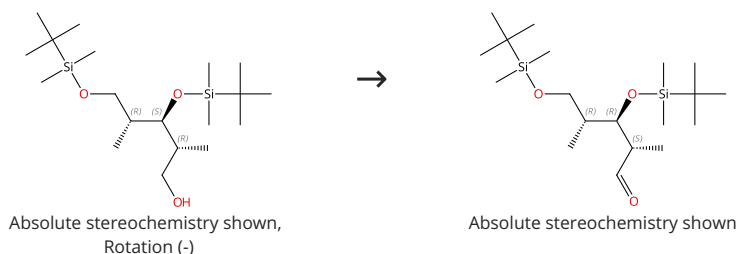
Effects of the ligand linkers on stability of mixed-valence Cu(I) Cu(II) and catalytic aerobic alcohol oxidation activity

By: Jehdaramarn, Attawit; et al

Scientific Reports (2024), 14(1), 15497.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-38407793

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Sodium bicarbonate, Dess-Martin periodinane
Solvents: Dichloromethane; 0 °C; 1 h, rt
- 1.2 **Reagents:** Sodium bicarbonate, Sodium thiosulfate
Solvents: Water; 0 °C

Experimental Protocols

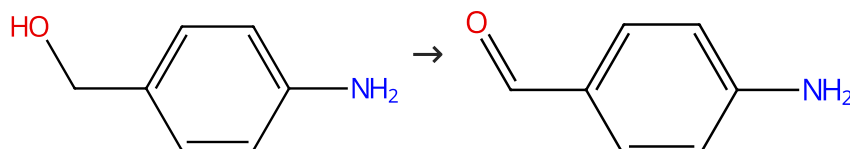
Total Synthesis of Calcaripeptide C

By: Patil, Shashikant; et al

European Journal of Organic Chemistry (2022), 2022(47), e202201078.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (98)

Suppliers (52)

31-480-CAS-22397501

Steps: 1 Yield: 100%

- 1.1 **Reagents:** 1-Methylimidazole, Oxygen
Catalysts: Tempo, 2445364-42-9
Solvents: Acetonitrile; 2 h, rt

Experimental Protocols

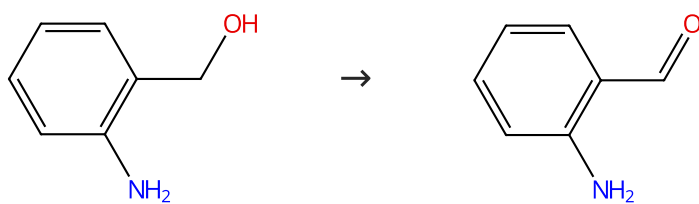
Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air

By: Lai, Ya-Liang; et al

Dalton Transactions (2020), 49(22), 7304-7308.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (105)

Suppliers (66)

31-480-CAS-22397499

Steps: 1 Yield: 100%

Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air

By: Lai, Ya-Liang; et al

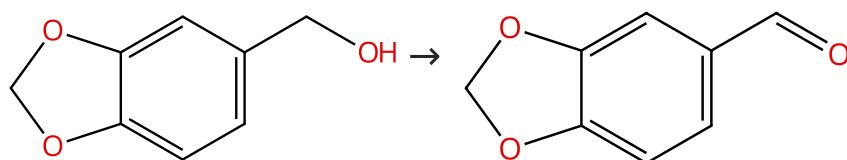
Dalton Transactions (2020), 49(22), 7304-7308.

1.1 **Reagents:** 1-Methylimidazole, Oxygen
Catalysts: Tempo, 2445364-42-9
Solvents: Acetonitrile; 2 h, rt

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (80)

Suppliers (42)

31-614-CAS-31479734

Steps: 1 Yield: 100%

Aerobic oxidation of alcohols enabled by nitrogen- doped copper nanoparticle catalysts

By: Tobita, Fumiya; et al

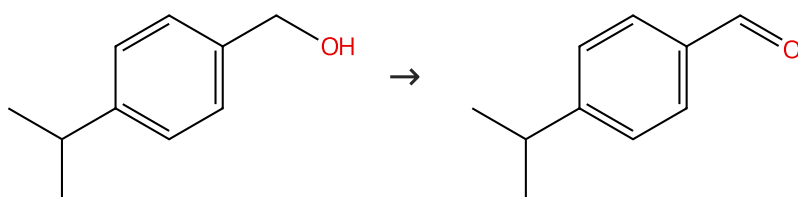
Catalysis Science & Technology (2022), 12(4), 1043-1048.

1.1 **Reagents:** Oxygen
Catalysts: 1-Methylimidazole, Tempo, Copper (nitrogen-doped carbon-incarcerated copper nanoparticle)
Solvents: Acetonitrile; 20 h, 1 bar, 60 °C

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (97)

Suppliers (92)

31-614-CAS-31479720

Steps: 1 Yield: 100%

Aerobic oxidation of alcohols enabled by nitrogen- doped copper nanoparticle catalysts

By: Tobita, Fumiya; et al

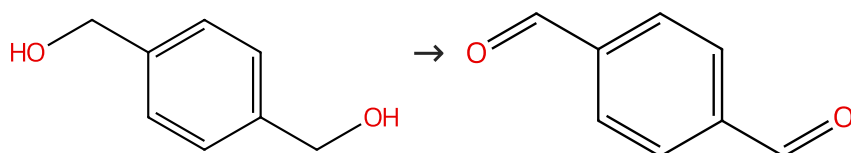
Catalysis Science & Technology (2022), 12(4), 1043-1048.

1.1 **Reagents:** Oxygen
Catalysts: 1-Methylimidazole, Tempo, Copper (nitrogen-doped carbon-incarcerated copper nanoparticle)
Solvents: Acetonitrile; 20 h, 1 bar, 60 °C

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (89)

Suppliers (83)

31-480-CAS-22397509

Steps: 1 Yield: 100%

Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air

By: Lai, Ya-Liang; et al

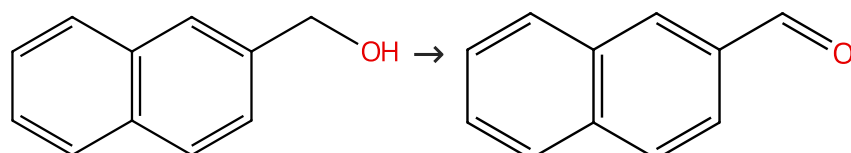
Dalton Transactions (2020), 49(22), 7304-7308.

1.1 **Reagents:** 1-Methylimidazole, Oxygen
Catalysts: Tempo, 2445364-42-9
Solvents: Acetonitrile; 2 h, rt

Experimental Protocols

Scheme 14 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (90)

Suppliers (86)

31-614-CAS-31479732

Steps: 1 Yield: 100%

Aerobic oxidation of alcohols enabled by nitrogen- doped copper nanoparticle catalysts

By: Tobita, Fumiya; et al

Catalysis Science & Technology (2022), 12(4), 1043-1048.

1.1 **Reagents:** Oxygen
Catalysts: 1-Methylimidazole, Tempo, Copper (nitrogen-doped carbon-incarcerated copper nanoparticle)
Solvents: Acetonitrile; 20 h, 1 bar, 60 °C

Experimental Protocols

31-480-CAS-22397508

Steps: 1 Yield: 100%

Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air

By: Lai, Ya-Liang; et al

Dalton Transactions (2020), 49(22), 7304-7308.

1.1 **Reagents:** 1-Methylimidazole, Oxygen
Catalysts: Tempo, 2445364-42-9
Solvents: Acetonitrile; 2 h, rt

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



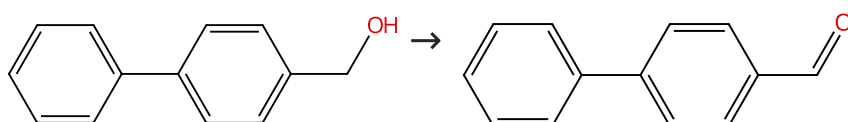
Suppliers (78)

Suppliers (85)

31-614-CAS-41257725	Steps: 1 Yield: 100%	Effects of the ligand linkers on stability of mixed-valence Cu(I) Cu(II) and catalytic aerobic alcohol oxidation activity
1.1 Catalysts: Copper bromide, 1,5-Naphthalenediamine, N^1,N^1,N^5,N^5 -tetrakis[[1-(phenylmethyl)-1 <i>H</i> -1,2,3-triazol-4-yl]methyl]- Solvents: Acetonitrile; 15 min, rt		By: Jehdaramarn, Attawit; et al
1.2 Catalysts: 1-Methylimidazole, Tempo Solvents: Acetonitrile; 2 h, rt		Scientific Reports (2024), 14(1), 15497.
Experimental Protocols		

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



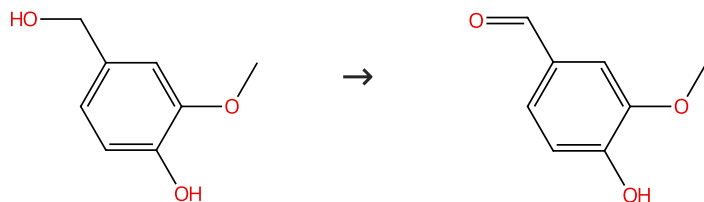
Suppliers (85)

Suppliers (100)

31-480-CAS-22397510	Steps: 1 Yield: 100%	Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air
1.1 Reagents: 1-Methylimidazole, Oxygen Catalysts: Tempo, 2445364-42-9 Solvents: Acetonitrile; 2 h, rt		By: Lai, Ya-Liang; et al
Experimental Protocols		Dalton Transactions (2020), 49(22), 7304-7308.

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



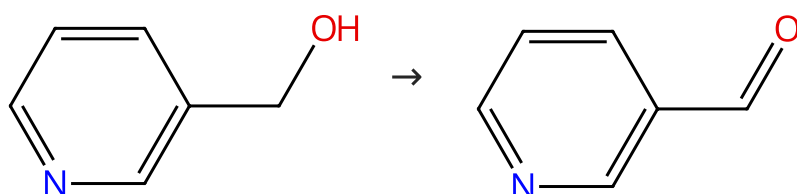
Suppliers (98)

Suppliers (145)

31-614-CAS-31870832	Steps: 1 Yield: 100%	Aerobic oxidation of vanillyl alcohol to vanillin catalyzed by air-stable and recyclable copper complex and TEMPO under base-free conditions
1.1 Reagents: Oxygen Catalysts: Tempo, 2763834-01-9 Solvents: Acetone, Water; 12 h, 40 °C		By: Jana, Narayan Ch.; et al
Experimental Protocols		Green Chemistry (2022), 24(6), 2542-2556.

Scheme 18 (2 Reactions)

Steps: 1 Yield: 100%



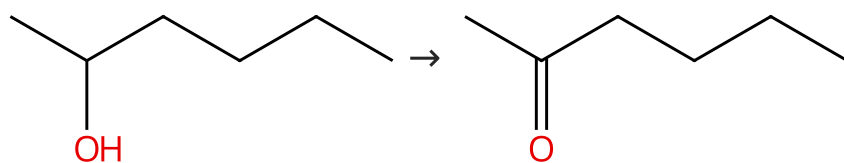
Suppliers (87)

Suppliers (92)

31-614-CAS-43504983 Steps: 1 Yield: 100% 1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene Catalysts: Tempo, Bipyridine, Copper(II) tetrafluoroborate Solvents: Dimethyl sulfoxide; 2 h, rt	An automatic end-to-end chemical synthesis development platform powered by large language models By: Ruan, Yixiang; et al Nature Communications (2024), 15(1), 10160.
31-480-CAS-22397505 Steps: 1 Yield: 100% 1.1 Reagents: 1-Methylimidazole, Oxygen Catalysts: Tempo, 2445364-42-9 Solvents: Acetonitrile; 2 h, rt Experimental Protocols	Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air By: Lai, Ya-Liang; et al Dalton Transactions (2020), 49(22), 7304-7308.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



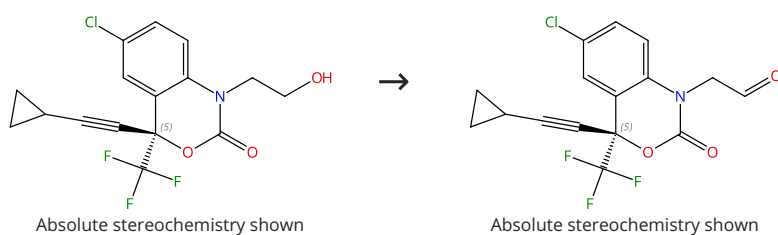
Suppliers (53)

Suppliers (70)

31-614-CAS-34026214 Steps: 1 Yield: 100% 1.1 Reagents: Oxygen Catalysts: Tempo, Silica, Ruthenium trichloride Solvents: Chlorobenzene; 6 h, 120 °C Experimental Protocols	TEMPO-Ru-BEA Composite Material for the Selective Oxidation of Alcohols to Aldehydes By: Deng, Jianying; et al ACS Catalysis (2022), 12(15), 8925-8935.
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Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



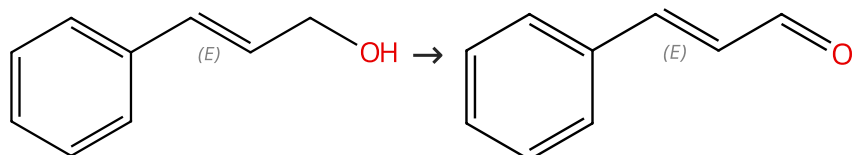
Absolute stereochemistry shown

Absolute stereochemistry shown

31-614-CAS-36103432 Steps: 1 Yield: 100% 1.1 Reagents: Dimethyl sulfoxide, Oxalyl chloride Solvents: Dichloromethane; 20 min, -60 °C 1.2 Solvents: Dichloromethane; -60 °C; 30 min, -60 °C 1.3 Reagents: Triethylamine; -60 °C → rt; 1 h, rt 1.4 Reagents: Ammonium chloride Solvents: Water; rt Experimental Protocols	Catalytic asymmetric C-C cross-couplings enabled by photoexcitation By: Crisenza, Giacomo E. M.; et al Nature Chemistry (2021), 13(6), 575-580.
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Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Double bond geometry shown

Suppliers (66)

Suppliers (109)

31-614-CAS-31479743

Steps: 1 Yield: 100%

Aerobic oxidation of alcohols enabled by nitrogen- doped copper nanoparticle catalysts

By: Tobita, Fumiya; et al

Catalysis Science & Technology (2022), 12(4), 1043-1048.

1.1 Reagents: Oxygen

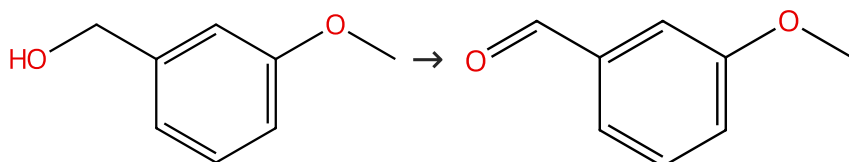
Catalysts: 1-Methylimidazole, Tempo, Copper (nitrogen-doped carbon-incarcerated copper nanoparticle)

Solvents: Acetonitrile; 20 h, 1 bar, 60 °C

Experimental Protocols

Scheme 22 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (77)

Suppliers (89)

31-614-CAS-41257716

Steps: 1 Yield: 100%

Effects of the ligand linkers on stability of mixed- valence Cu(I) Cu(II) and catalytic aerobic alcohol oxidation activity

By: Jehdaramarn, Attawit; et al

Scientific Reports (2024), 14(1), 15497.

1.1 Catalysts: Copper bromide, 1,5-Naphthalenediamine, N^1,N^1,N^5,N^5 -tetrakis[[1-(phenylmethyl)-1*H*-1,2,3-triazol-4-yl]methyl]-

Solvents: Acetonitrile; 15 min, rt

1.2 Catalysts: 1-Methylimidazole, Tempo

Solvents: Acetonitrile; 2 h, rt

Experimental Protocols

31-614-CAS-31479716

Steps: 1 Yield: 100%

Aerobic oxidation of alcohols enabled by nitrogen- doped copper nanoparticle catalysts

By: Tobita, Fumiya; et al

Catalysis Science & Technology (2022), 12(4), 1043-1048.

1.1 Reagents: Oxygen

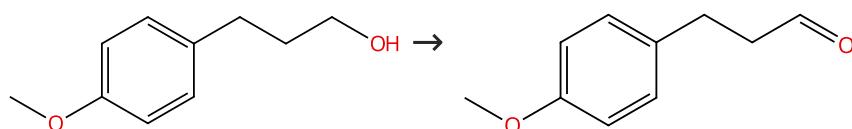
Catalysts: 1-Methylimidazole, Tempo, Copper (nitrogen-doped carbon-incarcerated copper nanoparticle)

Solvents: Acetonitrile; 20 h, 1 bar, 60 °C

Experimental Protocols

Scheme 23 (2 Reactions)

Steps: 1 Yield: 100%



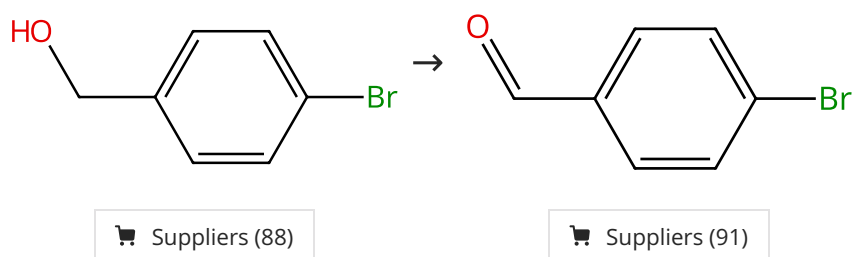
Suppliers (63)

Suppliers (57)

31-614-CAS-41260170 Steps: 1 Yield: 100% 1.1 Reagents: 2-Iodoxybenzoic acid Solvents: Ethyl acetate; 4 h, rt → reflux Experimental Protocols	Halogen-Bond-Promoted Direct Cross-Coupling of Trifluoromethylated Alkyl Bromides with Coumarins/Quinolinones: Unraveling Trifluoromethyl Effects By: Guo, Peng; et al Organic Letters (2024), 26(15), 3097-3102.
31-614-CAS-41257863 Steps: 1 Yield: 100% 1.1 Reagents: 2-Iodoxybenzoic acid Solvents: Ethyl acetate; 4 h, rt → reflux Experimental Protocols	Catalytic reduction of trifluoromethylated alkyl bromides and synthesis of alkylated heterocycles under visible light irradiation: synergetic action of a halogen bond and Ni catalysis By: Wang, Gai-Rong; et al Organic Chemistry Frontiers (2024), 11(17), 4722-4729.

Scheme 24 (3 Reactions)

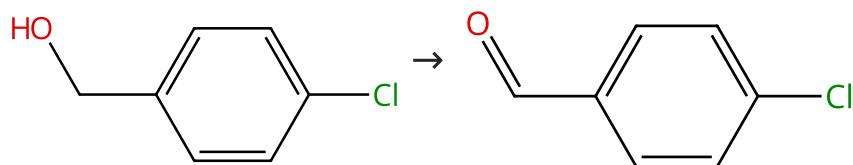
Steps: 1 Yield: 100%



31-614-CAS-41257722 Steps: 1 Yield: 100% 1.1 Catalysts: Copper bromide, 1,5-Naphthalenediamine, <i>N</i>¹,<i>N</i>¹,<i>N</i>⁵,<i>N</i>⁵-tetrakis[[1-(phenylmethyl)-1<i>H</i>-1,2,3-triazol-4-yl]methyl]- Solvents: Acetonitrile; 15 min, rt 1.2 Catalysts: 1-Methylimidazole, Tempo Solvents: Acetonitrile; 2 h, rt Experimental Protocols	Effects of the ligand linkers on stability of mixed-valence Cu(I) Cu(II) and catalytic aerobic alcohol oxidation activity By: Jehdaramarn, Attawit; et al Scientific Reports (2024), 14(1), 15497.
31-614-CAS-31479713 Steps: 1 Yield: 100% 1.1 Reagents: Oxygen Catalysts: 1-Methylimidazole, Tempo, Copper (nitrogen-doped carbon-incarcerated copper nanoparticle) Solvents: Acetonitrile; 20 h, 1 bar, 60 °C Experimental Protocols	Aerobic oxidation of alcohols enabled by nitrogen- doped copper nanoparticle catalysts By: Tobita, Fumiya; et al Catalysis Science & Technology (2022), 12(4), 1043-1048.
31-480-CAS-22397500 Steps: 1 Yield: 100% 1.1 Reagents: 1-Methylimidazole, Oxygen Catalysts: Tempo, 2445364-42-9 Solvents: Acetonitrile; 2 h, rt Experimental Protocols	Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air By: Lai, Ya-Liang; et al Dalton Transactions (2020), 49(22), 7304-7308.

Scheme 25 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (85)

Suppliers (98)

31-614-CAS-31479722

Steps: 1 Yield: 100%

Aerobic oxidation of alcohols enabled by nitrogen- doped copper nanoparticle catalysts

By: Tobita, Fumiya; et al

Catalysis Science & Technology (2022), 12(4), 1043-1048.

1.1 Reagents: Oxygen

Catalysts: 1-Methylimidazole, Tempo, Copper (nitrogen-doped carbon-incarcerated copper nanoparticle)

Solvents: Acetonitrile; 20 h, 1 bar, 60 °C

Experimental Protocols

31-480-CAS-22397498

Steps: 1 Yield: 100%

Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air

By: Lai, Ya-Liang; et al

Dalton Transactions (2020), 49(22), 7304-7308.

1.1 Reagents: 1-Methylimidazole, Oxygen

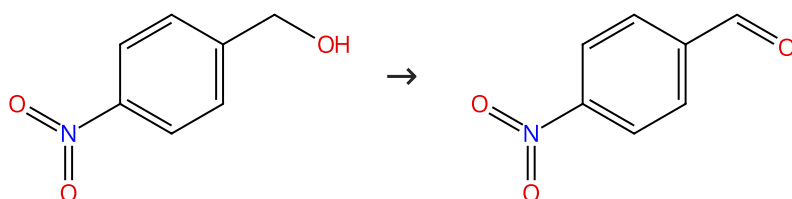
Catalysts: Tempo, 2445364-42-9

Solvents: Acetonitrile; 2 h, rt

Experimental Protocols

Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (96)

Suppliers (92)

31-480-CAS-22397496

Steps: 1 Yield: 100%

Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air

By: Lai, Ya-Liang; et al

Dalton Transactions (2020), 49(22), 7304-7308.

1.1 Reagents: 1-Methylimidazole, Oxygen

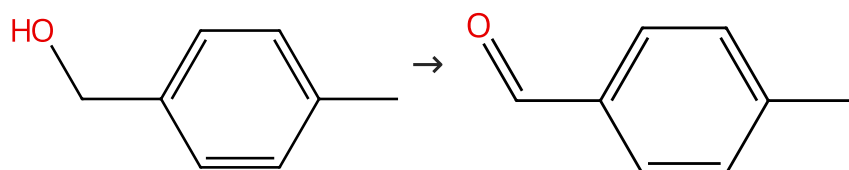
Catalysts: Tempo, 2445364-42-9

Solvents: Acetonitrile; 2 h, rt

Experimental Protocols

Scheme 27 (3 Reactions)

Steps: 1 Yield: 100%



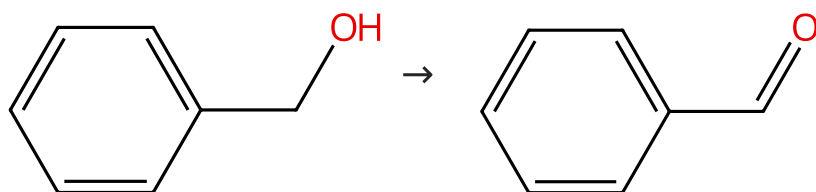
Suppliers (79)

Suppliers (104)

31-614-CAS-34226387 Steps: 1 Yield: 100% 1.1 Reagents: Oxygen Catalysts: Iron oxide (Fe₃O₄), Tempo, Silica Solvents: Toluene; 18 h, 1 atm, 50 °C Experimental Protocols	Selective oxidation of 5-hydroxymethylfurfural into 2,5-diformylfuran by TEMPO-assisted magnetic Fe₃O₄@SiO₂@mSiO₂-NH₂-Cu(II) catalytic system By: Pan, Pan; et al Molecular Catalysis (2022), 530, 112622.
31-614-CAS-31479728 Steps: 1 Yield: 100% 1.1 Reagents: Oxygen Catalysts: 1-Methylimidazole, Tempo, Copper (nitrogen-doped carbon-incarcerated copper nanoparticle) Solvents: Acetonitrile; 20 h, 1 bar, 60 °C Experimental Protocols	Aerobic oxidation of alcohols enabled by nitrogen-doped copper nanoparticle catalysts By: Tobita, Fumiya; et al Catalysis Science & Technology (2022), 12(4), 1043-1048.
31-480-CAS-22397497 Steps: 1 Yield: 100% 1.1 Reagents: 1-Methylimidazole, Oxygen Catalysts: Tempo, 2445364-42-9 Solvents: Acetonitrile; 2 h, rt Experimental Protocols	Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air By: Lai, Ya-Liang; et al Dalton Transactions (2020), 49(22), 7304-7308.

Scheme 28 (3 Reactions)

Steps: 1 Yield: 100%



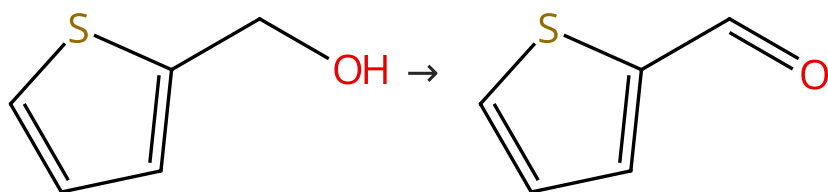
Suppliers (149)

Suppliers (69)

31-614-CAS-47238170 Steps: 1 Yield: 100% 1.1 Reagents: 1-Methylimidazole Catalysts: Tempo, Bipyridine, Copper(1+), tetrakis(acetonitrile)-, (7-4)-, 1,1,1-trifluoromethanesulfonate (1:1) Solvents: Acetonitrile; 3 h, 25 °C Experimental Protocols	On-Demand Electrochemical Synthesis of Tetrakisacetonitrile Copper(I) Triflate and Its Application in the Aerobic Oxidation of Alcohols By: Nicholls, Thomas P.; et al Inorganic Chemistry (2021), 60(10), 6976-6980.
31-480-CAS-22397495 Steps: 1 Yield: 100% 1.1 Reagents: 1-Methylimidazole, Oxygen Catalysts: Tempo, 2445364-42-9 Solvents: Acetonitrile; 2 h, rt Experimental Protocols	Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air By: Lai, Ya-Liang; et al Dalton Transactions (2020), 49(22), 7304-7308.
31-480-CAS-22397511 Steps: 1 Yield: 100% 1.1 Reagents: 1-Methylimidazole, Oxygen Catalysts: Tempo, Copper, tris[μ₃-(N¹,N³-bis[(1<i>H</i>-imidazol-5-yl)-κN¹:κN³]methylene]-1,3-propanediaminato(2-)-κN¹,κN²]]tris(perchlorato-κO)hexa-, stereoisomer Solvents: Acetonitrile; 2 h, rt Experimental Protocols	Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air By: Lai, Ya-Liang; et al Dalton Transactions (2020), 49(22), 7304-7308.

Scheme 29 (3 Reactions)

Steps: 1 Yield: 100%



Suppliers (81)

Suppliers (98)

31-614-CAS-43504987

Steps: 1 Yield: 100%

An automatic end-to-end chemical synthesis development platform powered by large language models

1.1 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene
Catalysts: Tempo, Copper(II) triflate, Bipyridine
Solvents: Dimethyl sulfoxide; 2 h, rt

By: Ruan, Yixiang; et al

Nature Communications (2024), 15(1), 10160.

31-614-CAS-31479737

Steps: 1 Yield: 100%

Aerobic oxidation of alcohols enabled by nitrogen- doped copper nanoparticle catalysts

1.1 **Reagents:** Oxygen
Catalysts: 1-Methylimidazole, Tempo, Copper (nitrogen-doped carbon-incarcerated copper nanoparticle)
Solvents: Acetonitrile; 20 h, 1 bar, 60 °C

By: Tobita, Fumiya; et al

Catalysis Science & Technology (2022), 12(4), 1043-1048.

Experimental Protocols

31-480-CAS-22397504

Steps: 1 Yield: 100%

Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air

1.1 **Reagents:** 1-Methylimidazole, Oxygen
Catalysts: Tempo, 2445364-42-9
Solvents: Acetonitrile; 2 h, rt

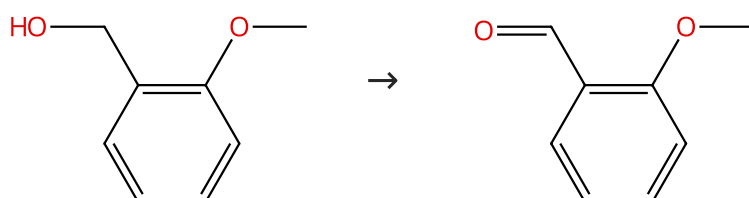
By: Lai, Ya-Liang; et al

Dalton Transactions (2020), 49(22), 7304-7308.

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (96)

31-614-CAS-41257715

Steps: 1 Yield: 100%

Effects of the ligand linkers on stability of mixed- valence Cu(I) Cu(II) and catalytic aerobic alcohol oxidation activity

1.1 **Catalysts:** Copper bromide, 1,5-Naphthalenediamine, N^1, N^1, N^5, N^5 -tetrakis[[1-(phenylmethyl)-1H-1,2,3-triazol-4-yl]methyl]-
Solvents: Acetonitrile; 15 min, rt

By: Jehdaramarn, Attawit; et al

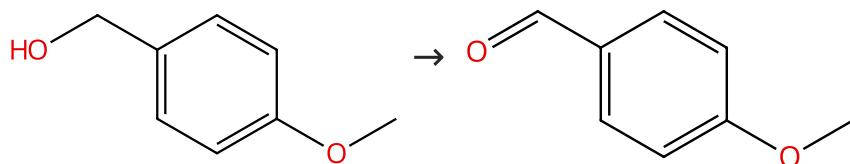
Scientific Reports (2024), 14(1), 15497.

1.2 **Catalysts:** 1-Methylimidazole, Tempo
Solvents: Acetonitrile; 2 h, rt

Experimental Protocols

Scheme 31 (5 Reactions)

Steps: 1 Yield: 100%



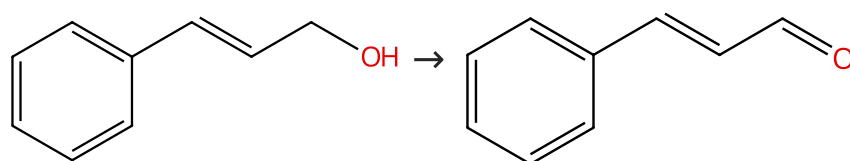
Suppliers (105)

Suppliers (128)

31-614-CAS-43504984 Steps: 1 Yield: 100% 1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene Catalysts: Tempo, Copper(II) triflate, Bipyridine Solvents: Dimethyl sulfoxide; 2 h, rt	An automatic end-to-end chemical synthesis development platform powered by large language models By: Ruan, Yixiang; et al Nature Communications (2024), 15(1), 10160.
31-614-CAS-41257718 Steps: 1 Yield: 100% 1.1 Catalysts: Copper bromide, 1,5-Naphthalenediamine, N^1, N^1, N^5, N^5-tetrakis[[1-(phenylmethyl)-1<i>H</i>-1,2,3-triazol-4-yl]methyl]- Solvents: Acetonitrile; 15 min, rt 1.2 Catalysts: 1-Methylimidazole, Tempo Solvents: Acetonitrile; 2 h, rt Experimental Protocols	Effects of the ligand linkers on stability of mixed-valence Cu(I) Cu(II) and catalytic aerobic alcohol oxidation activity By: Jehdaramarn, Attawit; et al Scientific Reports (2024), 14(1), 15497.
31-614-CAS-34226389 Steps: 1 Yield: 100% 1.1 Reagents: Oxygen Catalysts: Iron oxide (Fe_3O_4), Tempo, Silica Solvents: Toluene; 18 h, 1 atm, 50 °C Experimental Protocols	Selective oxidation of 5-hydroxymethylfurfural into 2,5-diformylfuran by TEMPO-assisted magnetic $\text{Fe}_3\text{O}_4@\text{SiO}_2@\text{mSiO}_2\text{-NH}_2\text{-Cu(II)}$ catalytic system By: Pan, Pan; et al Molecular Catalysis (2022), 530, 112622.
31-614-CAS-31479738 Steps: 1 Yield: 100% 1.1 Reagents: Oxygen Catalysts: 1-Methylimidazole, Tempo, Copper (nitrogen-doped carbon-incarcerated copper nanoparticle) Solvents: Acetonitrile; 20 h, 1 bar, 60 °C Experimental Protocols	Aerobic oxidation of alcohols enabled by nitrogen-doped copper nanoparticle catalysts By: Tobita, Fumiya; et al Catalysis Science & Technology (2022), 12(4), 1043-1048.
31-480-CAS-22397502 Steps: 1 Yield: 100% 1.1 Reagents: 1-Methylimidazole, Oxygen Catalysts: Tempo, 2445364-42-9 Solvents: Acetonitrile; 2 h, rt Experimental Protocols	Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air By: Lai, Ya-Liang; et al Dalton Transactions (2020), 49(22), 7304-7308.

Scheme 32 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (52)

31-614-CAS-43504986		Steps: 1 Yield: 100%	An automatic end-to-end chemical synthesis development platform powered by large language models By: Ruan, Yixiang; et al Nature Communications (2024), 15(1), 10160.
1.1	Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene Catalysts: Tempo, Bipyridine, Copper(II) tetrafluoroborate Solvents: Dimethyl sulfoxide; 2 h, rt		
31-480-CAS-22397506		Steps: 1 Yield: 100%	Self-assembly of mixed-valence and heterometallic metallo cycles: Efficient catalysts for the oxidation of alcohols to aldehydes in ambient air By: Lai, Ya-Liang; et al Dalton Transactions (2020), 49(22), 7304-7308.
1.1	Reagents: 1-Methylimidazole, Oxygen Catalysts: Tempo, 2445364-42-9 Solvents: Acetonitrile; 2 h, rt		
Experimental Protocols			